

Matthew Martin

Phone: (301)-820-5744 — Email: matthewmartin117@gmail.com

Portfolio: <https://matthewmartin117.github.io/>

Summary: Computer Science graduate with strong experience in **AI/ML**, **healthcare technology**, and **full-stack development**. Experienced in building **secure healthcare APIs**, **machine learning pipelines for medical imaging**, and **full-stack applications**. Passionate about leveraging **AI**, **NLP**, and **cloud technologies** to solve real-world healthcare challenges.

Education

University of Colorado Boulder — B.S. **Computer Science**, GPA: 4.0/4.0, Expected Dec 2025

James Madison University — B.S. **Biology**, Graduated Jul 2021

Technical Skills

Programming: Python, Java, SQL, JavaScript/React, Flask, Spring Boot, Docker, AWS, Git/GitHub

AI/ML & Healthcare Tech: PyTorch, TensorFlow/Keras, HuggingFace, NLP, CNNs, ML Pipelines

Data & Systems: PostgreSQL, ETL pipelines, HIPAA-compliant data handling

Work Experience

Data Engineer Intern — **AstraZeneca**, May 2025 – Aug 2025

- Built and maintained **ETL pipelines** for experimental datasets using SQL Server.
- Developed a C#/.NET application to automate data ingestion, improving pipeline accuracy and reliability.
- Collaborated with data scientists in **Agile sprints**, delivering demos for predictive modeling workflows.

Sampling Coordinator — **United States Pharmacopeia**, Mar 2022 – Jan 2023

- Managed pharmaceutical sample logging and shipments via **Oracle LIMS**, ensuring regulatory compliance.

Clinical Research Scientist — **Curative Health**, Aug 2021 – Dec 2021

- Processed large-scale **COVID-19 PCR datasets** with high **data integrity**.
 - Developed methods to ensure accuracy and compliance in **clinical data pipelines**.
-

Projects

Java RESTful API for Healthcare Data — **Java**, **Spring Boot**, **PostgreSQL**, **Docker**, **AWS**

- Designed and deployed secure **API** for sensitive biological sample data with **authentication and validation**.
- Built **AI/NLP microservice** to identify and redact **PHI** in patient records for **HIPAA compliance**.
- Implemented **controller-level CRUD operations**, maintaining database integrity.

HiFive Social Media App — **Python**, **Flask**, **React**, **PostgreSQL**

- Developed full-stack social web application with dynamic search and user matching.
- Integrated frontend with backend and local JSON for testing.

Pneumonia Detection from Chest X-Rays — **Python**, **TensorFlow**, **Keras**

- Developed **CNN and transfer learning models** achieving 0.92 recall.
- Performed preprocessing, class weighting, and augmentation for robust diagnostic support.